[INCH-POUND]

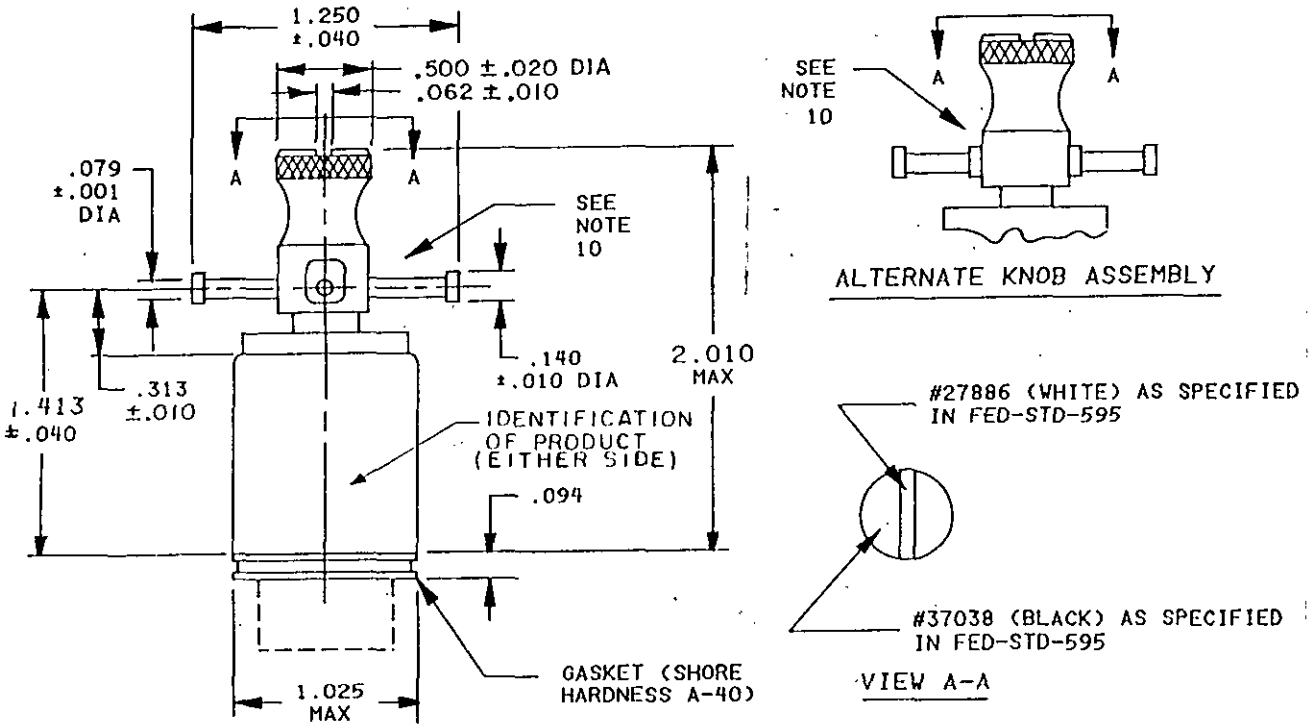
MIL-R-6106/1F(USAF)
28 March 1990
SUPERSEDING
MIL-R-6106/1E(USAF)
17 February 1988

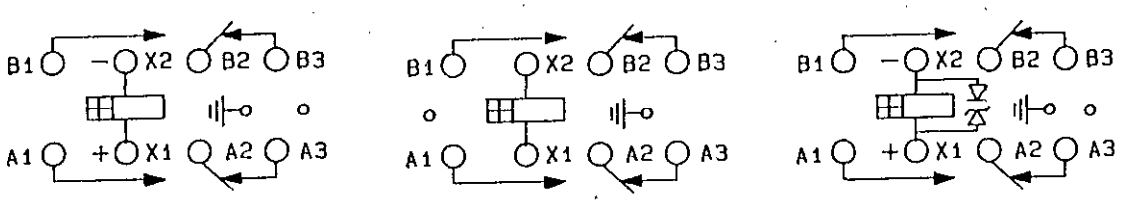
MILITARY SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, TYPE I, 2PDT, LOW LEVEL TO 10 AMPERES, PERMANENT MAGNET DRIVE, HERMETICALLY SEALED ASSEMBLY

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-R-6106.





(F) FIGURE 1. Relay, outline drawing.

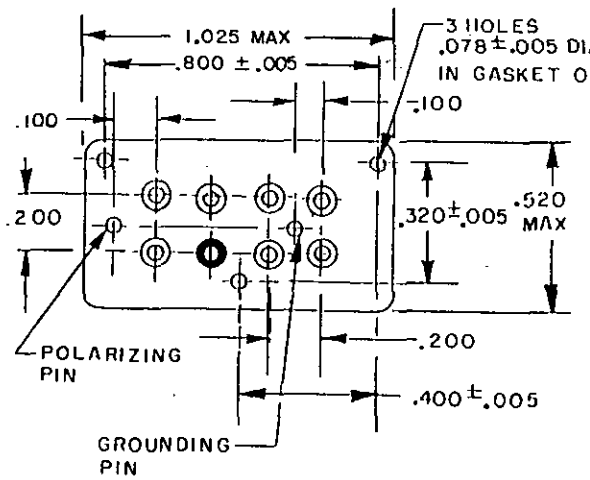
(F) denotes changes

FSC 5945

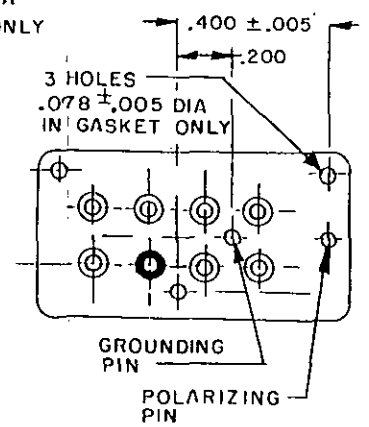
AMSC N/A

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

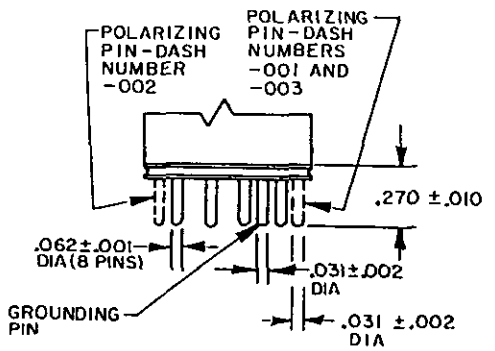
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TERMINAL LAYOUT - C (AC COILS)



TERMINAL LAYOUT - D (DC COILS)



TERMINALS (SEE NOTE 9)

Inches	mm	Inches	mm
.001	0.03	.140	3.56
.002	0.05	.200	5.08
.005	0.13	.270	6.86
.010	0.25	.313	7.95
.020	0.51	.320	8.13
.031	0.79	.400	10.16
.040	1.02	.520	13.21
.062	1.57	.800	20.32
.078	1.98	1.025	26.04
.079	2.01	1.250	31.75
.094	2.39	1.413	35.89
.100	2.54	2.010	51.05

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm).
4. Coil polarity not applicable to -002.
5. DC versions of this relay shall not operate or be damaged by reverse polarity. Semiconductors shall not be used for this purpose.
6. Permanent magnet drive consists of permanent magnet with its flux path switched and combined with the electromagnetic flux.
7. Terminal numbers shall not appear on the header. There shall be affixed to the relay a suitable legible circuit diagram that identifies each terminal location specified.
8. Back EMF suppression circuit: JAN TX or equivalent screened semiconductors shall be used. Back EMF shall be 42 V dc maximum. Relays using suppression devices shall continue to operate should the suppression circuit be in a failure mode.
9. All terminals shall be gold plated in accordance with MIL-G-45204, type II, class 1; underplating shall be nickel, 50 to 150 microinches thick.
10. Knob and post shall be nickel plated (100 microinches minimum) 303 stainless steel.

FIGURE 1. Relay, outline drawing - Continued.

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REQUIREMENTS:

Coil data: See table I.

Duty rating: Continuous.

Operational data:

Time-current relay characteristics: Applicable (see table II).

Rated contact load: See table III.

Operate time: See table I.

Release time: See table I.

Break bounce: See table I.

Physical:

Dimensions and configuration: See figure 1.

Weight (maximum): .120 pound (54.5 grams).

Terminal type: See figure 1.

Finish: Enclosure shall be in accordance with FED-STD-595:

-001 Black 37038
-002 Gray 36231
-003 Black 37038

Knob strength test: The rod to knob junction shall be capable of resisting an axial pull of 35 pounds without degradation.

Ⓕ Operational reliability: Applicable.

Ⓕ Identification of product: Applicable. The circuit diagram shall be oriented so that when the relay is held with the circuit diagram right side up as shown on figure 1, and rotated away from the viewer about a horizontal axis through the diagram until the header terminals face the viewer, then each terminal shall be in the location shown in the circuit diagram.

ENVIRONMENTAL CHARACTERISTICS:

Temperature rating: -70°C to +125°C.

Maximum altitude rating: 80,000 feet.

Ⓕ Shock: MIL-STD-202, method 213, test condition A 1/:

G-level: 50 g's.

Duration: 10 ms.

Maximum duration contact opening: 10 µs.

Maximum duration contact closing: 1 µs.

1/ To be tested as complete relay assembly (see table I, footnote 1/).

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Ⓕ TABLE I. Dash number and operating characteristics. 1/

Dash no.	Coil data: Nominal: Volts 2/	Nominal: Freq Hz	Nominal: Res Ω±10% @25°C	Max: Volts	Max: Ampere	Max pick-up voltage: Nominal 3/	Max pick-up voltage: High temp test	Max pick-up voltage: Cont current test	Hold voltage V dc 3/	Dropout voltage V dc 3/	Time-milliseconds - max: Operate 4/	Time-milliseconds - max: Release 5/	Bounce: Main NO	Bounce: Main NC	Bounce: Aux NO	Bounce: Aux NC	Overload current amperes	Rupture current amperes	Circuit schematic	Terminal layout
001	28 V dc	N/A	320	30	0.11	18.0	20	22.5	7.0	1.5	10	20	1	1	N/A	N/A	40	50	Fig 1A	Fig 1D
002	115 V ac	400	N/A	122	.04	90	96	103	30	5.0	15	50	1	1	N/A	N/A	60	80	Fig 1B	Fig 1C
003 6/	28 V dc	N/A	320	30	0.11	18.0	20	22.5	7.0	1.5	10	20	1	1	N/A	N/A	40	50	Fig 1C	Fig 1D

1/ The complete relay assembly shall consist of the following:

a. Required relay.

b. Relay, electric, bracket and socket assembly (see MIL-R-6106/4).

c. Relay electric, mounting track (see MIL-R-6106/3).

2/ CAUTION: The use of any coil voltage less than the nominal coil voltage will compromise the operation of the relay.

3/ Over temperature range.

4/ With nominal coil voltage.

5/ From nominal coil voltage.

6/ Dash number -003 equipped with back EMF (coil transient) suppression.

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TABLE II. Time-current relay characteristics at +25°C. 1/ 2/

1	15A - 1 hour
2	50A - 5.0 seconds
3	100A - 1.2 seconds
4	250A - 0.2 second
5	350A - 0.1 second

1/ Caution: Compare with time-current characteristics of the associated circuit protective device.

2/ Relays shall sustain five applications (make and carry only) of power concurrently on adjacent poles at each of five different current levels for the time durations in table II. Separate relays shall be tested at 28 V dc and 115/200 V ac, 400 Hz, 3 phase. Cooling time between successive applications shall be 30 minutes. The test shall be performed on both normally open and normally closed contacts of each relay. There shall be no failures or evidence of welding or sticking and relays shall pass contact voltage drop at conclusion.

Vibration, sinusoidal 2/:

G-level: 30 g's.

Frequency range: 10 to 3,000 Hz.

(F) Vibration scan: Applicable to group A testing.

Frequency range: 5 to 2000 Hz.

Intensity: 0.01 inch double amplitude or ±2 g's, whichever is less.

(F) Vibration, random 2/: MIL-STD-202, method 214, test condition IG ($0.4\ g^2/Hz$) 15 minutes each plane: 50 to 2,000 Hz.

Shock: Test condition C.

Acceleration 2/: 15 g's.

ELECTRICAL CHARACTERISTICS:

Insulation resistance, initial: 100 megohms.

After life or environmental tests: 100 megohms.

2/ To be tested as complete relay assembly (see table I, footnote 1/).

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Ⓕ TABLE III. Rated contact load (amperes per pole) (case grounded).

Type of load	Life operating cycles x 10^3	28 V dc				115 V ac, 1 phase				115/200 V ac, 3 phase 1/			
		Main		Aux		Main		Aux		Main		Aux	
		NO	NC	NO	NC	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz
Resistive	100	10	10	---	---	10	---	---	---	10	---	---	---
Inductive	20	8	8	---	---	8	---	---	---	8	---	---	---
Inductive						(Not applicable)							
Motor	100	4	4	---	---	4	---	---	---	4	---	---	---
Lamp	100	2	2	---	---	2	---	---	---	2	---	---	---
Transfer, load 2/													
Mechanical life (reduced load)	400	2.5	2.5	---	---	2.5	---	---	---	2.5	---	---	---
Low level	100			(Applicable per MIL-R-6106) 3/ 4/ 5/									
Intermediate current				(Applicable per MIL-R-6106)									

1/ 115/200 V ac for 60 Hz ratings, absence of value indicates relay is not rated for 3-phase applications.

2/ Transfer load indicates relay suitable for transfer between unsynchronized ac power supplies at rating indicated.

3/ For low level, the following shall apply:

a. During endurance and operational reliability testing, contact load shall be 10 to 50 microamperes; 10- to 50-millivolt open circuit voltage, 100-ohm maximum contact resistance.

b. Static contact resistance shall be performed at 50 mA maximum, 50 mV maximum.

4/ For group A contact voltage drop test, high level testing shall be performed first, followed by low level testing. The contacts shall not make or break the high level load.

5/ Relays shall be capable of switching 10 mA at 12.0 V dc.

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Dielectric withstanding voltage (sea level):

	Initial	After life tests
Coil to case	1,050	1,050
Aux contacts	---	---
All other points	1,500 3/	1,250

Dielectric withstanding voltage (altitude):

	Initial	After life tests
Coil to case	350	350
Aux contacts	---	---
All other points	350	350

Contact voltage drop or resistance:

High level:

Initial: 0.150 volt maximum.

After life: 0.175 volt maximum.

Low level:

Initial: 0.050 ohm maximum.

After life: 0.150 ohm maximum.

RFI standard (applicable to -002): MIL-STD-461.

Back EMF (coil transient): ±42 volts maximum.

(Applicable to -003)

Test temperature: Room ambient.

Point of measurement: Across coil terminals X_1 - X_2.

Coil voltage: 28 V dc.

Test method: Apply coil voltage to X_1 - X_2. With oscilloscope monitoring the coil terminals, measure and record the back electromotive force when voltage is removed.

Part or Identifying Number (PIN): M6106/1-(dash number from table I).

3/ To be tested with relay inserted in socket, MIL-R-6106/4.

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CONCLUDING MATERIAL

Custodian:
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

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